Results Appendix: Baseline vs. Re-Ranking RAG over the Constitution of India

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Code and data: https://github.com/Sahoo-Achyutananda/theRAGofakind

All values are computed from the saved runs in `failure_modes/results/` and the latency cells of `rag.ipynb`. **Baseline** = bi-encoder top 10 → top 3; **re-ranked** = the same 10 re-scored by the cross-encoder → top 3. Relevance is article-level unless stated: a chunk is relevant if its article is in the query's expected list.

- **First rank**: rank of the first relevant chunk in the top 10 (– = not in the top 10)
- **Recall@3**: share of the expected articles that appear in the top 3
- **Precision@3**: share of the 3 chunks sent to the LLM that are relevant
- **RR / MRR**: 1/rank of the first relevant chunk (0 if none); MRR is the mean over queries
- **nDCG@10**: every relevant article counts, discounted by $\log_2(\text{rank}+1)$, divided by the ideal ordering

Cells read baseline→re-ranked; ▲ / ▼ = better / worse after re-ranking.

A. Failure categories tested

Category	What it tests	Queries
Keyword trap / distractor	The query shares many words with the wrong passage: mostly near-duplicate Union vs. State articles (President vs. Governor, Parliament vs. State Legislature).	10
Synonyms / different terminology	The user's everyday words differ from the Constitution's legal wording ("remain silent" vs. "compelled to be a witness against himself").	10
Exceptions & corner cases	The general rule sits in one clause and the exception ("Nothing in this article shall…", "Provided that…") in another.	10
Negation & constraints	Questions about what is *not* allowed or who is excluded, where embeddings blur positive and negative statements.	7
Document-specific	Article-number lookups, omitted articles, cross-references ("the said clause") and section headings that are not part of the article text.	9
Out-of-corpus / hallucination checks	The answer is not in the article text (court doctrines, footnote-only amendment history, Schedules); the correct behaviour is to refuse.	6
Total		**52**

B. Summary: mean metrics per category

Only queries with an expected article are averaged (the four "refuse" queries are excluded).

Category	n	Recall@3	Precision@3	MRR	nDCG@10
Keyword trap / distractor	10	0.75→0.95 ▲	0.40→0.40	0.78→0.88 ▲	0.81→0.92 ▲
Synonyms / different terminology	10	0.50→0.70 ▲	0.27→0.37 ▲	0.54→0.62 ▲	0.56→0.63 ▲
Exceptions & corner cases	10	0.65→0.75 ▲	0.47→0.57 ▲	0.73→0.71 ▼	0.78→0.75 ▼
Negation & constraints	7	0.71→0.86 ▲	0.43→0.48 ▲	0.74→0.86 ▲	0.79→0.86 ▲
Document-specific	9	0.61→0.61	0.26→0.37 ▲	0.61→0.54 ▼	0.58→0.56 ▼
Out-of-corpus / hallucination checks	2	0.50→0.50	0.17→0.17	0.50→0.50	0.50→0.50
All	**48**	**0.64→0.76 ▲**	**0.35→0.42 ▲**	**0.67→0.71 ▲**	**0.69→0.73 ▲**

Hit rate	Baseline	Re-ranked
Hit@1	0.60	0.62
Hit@3	0.69	0.81
Hit@10	0.83	0.83

Hit@10 is identical by construction: re-ranking only reorders the same 10 chunks. After re-ranking, the first relevant chunk moved **up in 8**, **down in 8** and stayed in place in 24 queries; 8 queries had no relevant chunk in the top 10 (Section E).

C. Queries and per-query metrics

Keyword trap / distractor

Query	Expected	First rank	Recall@3	Precision@3	RR	nDCG@10
Can the Governor pardon a person sentenced to death?	72, 161	1→1	1.00→1.00	0.67→0.67	1.00→1.00	1.00→1.00
What is the minimum age to become President of India?	58	1→1	1.00→1.00	0.33→0.33	1.00→1.00	1.00→1.00
What is the minimum age to become a member of the Lok Sabha?	84	1→1	1.00→1.00	0.33→0.33	1.00→1.00	1.00→1.00
Who can issue Ordinances when the legislature is not in session?	123, 213	1→1	0.50→0.50	0.67→0.33 ▼	1.00→1.00	0.85→0.92 ▲
Which court can issue a writ of habeas corpus?	32, 226	1→2	1.00→1.00	0.67→0.67	1.00→0.50 ▼	0.92→0.69 ▼
What is the maximum gap allowed between two sessions of Parliament?	85	1→1	1.00→1.00	0.33→0.33	1.00→1.00	1.00→1.00
Can the President be a member of Parliament?	59	3→1	1.00→1.00	0.33→0.33	0.33→1.00 ▲	0.50→1.00 ▲
Who appoints the judges of a High Court?	217	5→1	0.00→1.00 ▲	0.00→0.33 ▲	0.20→1.00 ▲	0.39→1.00 ▲
Who is the supreme commander of the Defence Forces?	53	4→1	0.00→1.00 ▲	0.00→0.33 ▲	0.25→1.00 ▲	0.43→1.00 ▲
What is a Money Bill?	110	1→3	1.00→1.00	0.67→0.33 ▼	1.00→0.33 ▼	1.00→0.63 ▼

Synonyms / different terminology

Query	Expected	First rank	Recall@3	Precision@3	RR	nDCG@10
Does an accused person have the right to remain silent?	20	6→2	0.00→1.00 ▲	0.00→0.33 ▲	0.17→0.50 ▲	0.39→0.63 ▲
Can the government ban a peaceful protest?	19	–→–	0.00→0.00	0.00→0.00	0.00→0.00	0.00→0.00
Is child labour allowed in India?	24	1→1	1.00→1.00	0.33→0.33	1.00→1.00	1.00→1.00
Is human trafficking illegal?	23	1→1	1.00→1.00	0.33→0.33	1.00→1.00	1.00→1.00
Can the President be sued in court?	361	–→–	0.00→0.00	0.00→0.00	0.00→0.00	0.00→0.00
How can the President be removed from office?	56, 61	2→3	0.50→0.50	0.33→0.33	0.50→0.33 ▼	0.62→0.57 ▼
At what age can a citizen vote?	326	1→1	1.00→1.00	0.33→0.33	1.00→1.00	1.00→1.00
What is the national language of India?	343	2→3	1.00→1.00	0.33→0.33	0.50→0.33 ▼	0.63→0.50 ▼
Is caste discrimination illegal?	15, 17	1→1	0.50→0.50	1.00→1.00	1.00→1.00	0.61→0.61
Is the President immune from court proceedings?	361	5→1	0.00→1.00 ▲	0.00→0.67 ▲	0.20→1.00 ▲	0.39→1.00 ▲

Exceptions & corner cases

Query	Expected	First rank	Recall@3	Precision@3	RR	nDCG@10
Can the State make special laws only for women?	15	1→1	1.00→1.00	0.67→1.00 ▲	1.00→1.00	1.00→1.00
Can the State reserve government jobs for backward classes?	16	10→10	0.00→0.00	0.00→0.00	0.10→0.10	0.39→0.39
Can fundamental rights be suspended during an emergency?	359	1→1	1.00→1.00	1.00→1.00	1.00→1.00	1.00→1.00
Can the right to life be suspended during an emergency?	359, 21	1→1	0.50→0.50	0.67→1.00 ▲	1.00→1.00	0.92→0.88 ▼
Does Article 13 invalidate constitutional amendments that violate fundamental rights?	13	1→1	1.00→1.00	1.00→1.00	1.00→1.00	1.00→1.00
Can a person who is not an MP be appointed as a Union Minister?	75	1→2	1.00→1.00	0.33→0.33	1.00→0.50 ▼	1.00→0.63 ▼
Can a person who is not an MLA become a State Minister?	164	1→1	1.00→1.00	0.67→0.67	1.00→1.00	1.00→1.00
Can citizens be stopped from moving freely in Scheduled Tribe areas?	19	–→–	0.00→0.00	0.00→0.00	0.00→0.00	0.00→0.00
On what grounds can free speech be restricted?	19	1→2	1.00→1.00	0.33→0.33	1.00→0.50 ▼	1.00→0.63 ▼
Can the Vice-President act as President?	65	4→1	0.00→1.00 ▲	0.00→0.33 ▲	0.25→1.00 ▲	0.50→1.00 ▲

Negation & constraints

Query	Expected	First rank	Recall@3	Precision@3	RR	nDCG@10
Who is NOT protected by the arrest and detention safeguards?	22	1→1	1.00→1.00	1.00→1.00	1.00→1.00	1.00→1.00
Within how many hours must an arrested person be produced before a magistrate?	22	1→1	1.00→1.00	1.00→1.00	1.00→1.00	1.00→1.00
What can the Rajya Sabha NOT do with a Money Bill?	109	–→–	0.00→0.00	0.00→0.00	0.00→0.00	0.00→0.00
Which titles is the State NOT allowed to confer?	18	1→1	1.00→1.00	0.33→0.33	1.00→1.00	1.00→1.00
Can a non-citizen holding a public office accept a foreign title without permission?	18	1→1	1.00→1.00	0.33→0.33	1.00→1.00	1.00→1.00
Is religious instruction allowed in government-funded schools?	28	1→1	1.00→1.00	0.33→0.33	1.00→1.00	1.00→1.00
Can the Council of States be dissolved?	83	6→1	0.00→1.00 ▲	0.00→0.33 ▲	0.17→1.00 ▲	0.50→1.00 ▲

Document-specific

Query	Expected	First rank	Recall@3	Precision@3	RR	nDCG@10
What does Article 21 say?	21	1→1	1.00→1.00	0.33→0.33	1.00→1.00	1.00→1.00
What is Article 356 about?	356	1→1	1.00→1.00	0.67→1.00 ▲	1.00→1.00	1.00→1.00
What is Article 370?	370	2→1	1.00→1.00	0.33→0.67 ▲	0.50→1.00 ▲	0.63→1.00 ▲
Is the right to property a fundamental right?	300A	–→–	0.00→0.00	0.00→0.00	0.00→0.00	0.00→0.00
What does Article 31 say?	31	–→–	0.00→0.00	0.00→0.00	0.00→0.00	0.00→0.00
Can peaceful assembly be restricted in the interest of public order?	19	–→–	0.00→0.00	0.00→0.00	0.00→0.00	0.00→0.00
What does the right to equality say?	14	1→3	1.00→1.00	0.33→0.33	1.00→0.33 ▼	1.00→0.63 ▼
What are the fundamental duties of citizens?	51A	1→1	1.00→1.00	0.33→0.67 ▲	1.00→1.00	1.00→1.00
What are the Directive Principles of State Policy?	36, 37	1→2	0.50→0.50	0.33→0.33	1.00→0.50 ▼	0.61→0.39 ▼

Out-of-corpus / hallucination checks

Query	Expected	First rank	Recall@3	Precision@3	RR	nDCG@10
Is the right to privacy a fundamental right?	*refuse*	*no article to retrieve: judged on the answer*				
When was the right to education added to the Constitution?	21A	1→1	1.00→1.00	0.33→0.33	1.00→1.00	1.00→1.00
Which amendment inserted Article 21A?	21A	–→–	0.00→0.00	0.00→0.00	0.00→0.00	0.00→0.00
What is the basic structure doctrine?	*refuse*	*no article to retrieve: judged on the answer*				
What is the definition of an illiterate person in the Constitution?	*refuse*	*no article to retrieve: judged on the answer*				
What does the First Schedule list?	*refuse*	*no article to retrieve: judged on the answer*				

D. Selected rescue cases (clause-level relevance)

These cases were chosen *because* re-ranking promoted the answering clause, so their averages are biased in favour of re-ranking; they illustrate the mechanism rather than overall performance.

Query	Answering chunks (bi→ce rank)	Baseline top 3	Re-ranked top 3	Recall@3	RR	nDCG@10
What is Article 370?	370_0 8→2	232_0, 370_3, 145_2	370_3, 370_0, 344_5	0.00→1.00 ▲	0.12→0.50 ▲	0.32→0.63 ▲
Can the Vice-President act as President?	65_0 4→1	66_3, 71_0, 66_1	65_0, 64_0, 66_1	0.00→1.00 ▲	0.25→1.00 ▲	0.43→1.00 ▲
Who appoints the judges of a High Court?	217_0 7→1	233_0, 224A_0, 224_0	217_0, 216_0, 229_0	0.00→1.00 ▲	0.14→1.00 ▲	0.33→1.00 ▲
Who is the supreme commander of the Defence Forces?	53_0 4→1	148_0, 33_0, 76_0	53_0, 33_0, 146_0	0.00→1.00 ▲	0.25→1.00 ▲	0.43→1.00 ▲
Can the Council of States be dissolved?	83_0 6→1	172_0, 80_0, 80_1	83_0, 172_0, 90_0	0.00→1.00 ▲	0.17→1.00 ▲	0.36→1.00 ▲
Does an accused person have the right to remain silent?	20_0 6→2	311_1, 22_5, 22_0	22_0, 20_0, 22_4	0.00→1.00 ▲	0.17→0.50 ▲	0.36→0.63 ▲
Is caste discrimination illegal?	15_0 6→1, 15_1 10→3	15_3, 15_4, 15_6	15_0, 15_3, 15_1	0.00→1.00 ▲	0.17→1.00 ▲	0.40→0.92 ▲
Is the President immune from court proceedings?	361_1 7→1, 361_0 5→2	124_4, 128_0, 356_1	361_1, 361_0, 124_4	0.00→1.00 ▲	0.20→1.00 ▲	0.44→1.00 ▲
Mean				**0.00→1.00 ▲**	**0.18→0.88 ▲**	**0.38→0.90 ▲**

Answers (same prompt and model for both pipelines)

What is Article 370?

- *Answer 1 (baseline):* Based on the provided context, Article 370 deals with "Temporary provisions with respect to the State of Jammu and Kashmir" (Article 370). Under this article, the President may declare by public notification that the article shall cease to be operative or shall be operative only with specified exceptions and modifications from a specified date (Article 370). However, the recommendation of the Legislative Assembly of the State is necessary before the President issues such a notification (Article 370).
- *Answer 2 (re-ranked):* Based on the provided context, Article 370 consists of temporary provisions with respect to the State of Jammu and Kashmir (Article 370). Key provisions include: • **Non-applicability of Article 238:** The provisions of Article 238 do not apply in relation to the State of Jammu and Kashmir (Article 370). • **Limitation on Parliamentary Powers:** The power of Parliament to make laws for Jammu and Kashmir is limited to matters in the Union List and Concurrent List that are declared by the President, in consultation with or concurrence of the Government of the State, to correspond to the Instrument of Accession or as otherwise specified by order (Article 370). • **Power to Cease or Modify:** The President may declare by public notification that Article 370 shall cease to be operative or shall be operative with exceptions and modifications, provided that a recommendation from the Legislative Assembly of the State is necessary before issuing such a notification (Article 370).

Can the Vice-President act as President?

- *Answer 1 (baseline):* The provided context does not answer this.
- *Answer 2 (re-ranked):* Yes, the Vice-President can act as President. According to Article 65(1), the Vice-President shall act as President in the event of any vacancy in the office of the President occurring by reason of death, resignation, removal, or otherwise, until a new President is elected and enters upon the office. Additionally, under Article 65(2), when the President is unable to discharge their functions owing to absence, illness, or any other cause, the Vice-President shall discharge those functions until the President resumes duties.

Who appoints the judges of a High Court?

- *Answer 1 (baseline):* Based on the provided context, the appointment of regular permanent High Court judges is not mentioned. However, for specific types of High Court judges: • **Additional and Acting Judges:** They are appointed by the President, in consultation with the National Judicial Appointments Commission [Article 224(1), Article 224(2)]. • **Retired Judges (sitting at High Court sittings):** They are requested to sit and act by the National Judicial Appointments Commission (on a reference by the Chief Justice of the High Court) with the previous consent of the President [Article 224A]. If the question refers to regular permanent High Court judges, the provided context does not answer this.
- *Answer 2 (re-ranked):* Based on the provided context, the judges of a High Court are appointed by the President (Article 217, Article 216). Specifically: • Article 217(1) states that every Judge of a High Court shall be appointed by the President by warrant under his hand and seal. • Article 216 states that a High Court consists of a Chief Justice and such other Judges as the President may from time to time deem necessary to appoint.

Who is the supreme commander of the Defence Forces?

- *Answer 1 (baseline):* The provided context does not answer this.
- *Answer 2 (re-ranked):* Based on the provided context, the supreme command of the Defence Forces of the Union is vested in the President (Article 53).

Can the Council of States be dissolved?

- *Answer 1 (baseline):* The provided context does not answer this.
- *Answer 2 (re-ranked):* No, the Council of States cannot be dissolved. According to Article 83(1), the Council of States shall not be subject to dissolution, but as nearly as possible one-third of its members shall retire on the expiration of every second year.

Does an accused person have the right to remain silent?

- *Answer 1 (baseline):* The provided context does not answer this.
- *Answer 2 (re-ranked):* The provided context does not answer this. (Note: Under Article 20(3), the context states that no person accused of any offence shall be compelled to be a witness against himself, but it does not explicitly mention a "right to remain silent.")

Is caste discrimination illegal?

- *Answer 1 (baseline):* Yes, discrimination on the grounds of caste is prohibited under Article 15 [Article 15]. However, the State is not prevented from making special provisions for the advancement of socially and educationally backward classes of citizens, the Scheduled Castes, or the Scheduled Tribes, including special provisions regarding admission to educational institutions [Article 15(4), Article 15(5)].
- *Answer 2 (re-ranked):* Yes, caste discrimination is prohibited under Article 15: • **Discrimination by the State:** The State shall not discriminate against any citizen on grounds only of caste (among other grounds) under Article 15(1). • **Restrictions on citizens:** No citizen can be subjected to any disability, liability, restriction, or condition based on caste regarding access to shops, public restaurants, hotels, and places of public entertainment, or the use of wells, tanks, bathing ghats, roads, and places of public resort maintained out of State funds or dedicated to the general public under Article 15(2).

Is the President immune from court proceedings?

- *Answer 1 (baseline):* The provided context does not answer this.
- *Answer 2 (re-ranked):* Yes, the President enjoys immunity from certain court proceedings as specified under Article 361: • **Immunity for official acts:** Under Article 361(1), the President is not answerable to any court for the exercise and performance of the powers and duties of his office, or for any act done or purporting to be done by him in the exercise and performance of those powers and duties. • **Exceptions/Qualifications under Article 361(1):** • The conduct of the President may be brought under review by any court, tribunal, or body appointed or designated by either House of Parliament for the investigation of a charge under Article 61 (Article 361(1)). • This immunity does not restrict the right of any person to bring appropriate proceedings against the Government of India (Article 361(1)). • **Immunity from criminal proceedings:** Under Article 361(2), no criminal proceedings whatsoever can be instituted or continued against the President in any court during his term of office.

E. Queries with no relevant chunk in the top 10

Re-ranking cannot help these: the cross-encoder only re-orders the bi-encoder's top 10. *Actual rank* is the first relevant chunk's position when all 1,035 chunks are ranked.

Category	Query	Expected	Actual rank	Cosine (vs. #10)	Retrieved instead (top 3)
Synonyms / different terminology	Can the government ban a peaceful protest?	19	12	0.327 vs. 0.330	359_3, 359_0, 123_2
Synonyms / different terminology	Can the President be sued in court?	361	11	0.433 vs. 0.438	356_1, 224A_0, 222_0
Exceptions & corner cases	Can citizens be stopped from moving freely in Scheduled Tribe areas?	19	29	0.358 vs. 0.404	342_0, 366_25, 244_0
Negation & constraints	What can the Rajya Sabha NOT do with a Money Bill?	109	12	0.543 vs. 0.549	117_2, 110_0, 198_0
Document-specific	Is the right to property a fundamental right?	300A	15	0.389 vs. 0.408	31_0, 13_3, 31A_0
Document-specific	What does Article 31 say?	31	15	0.540 vs. 0.549	31D_0, 32A_0, 31B_0
Document-specific	Can peaceful assembly be restricted in the interest of public order?	19	12	0.394 vs. 0.395	359_5, 33_0, 121_0
Out-of-corpus / hallucination checks	Which amendment inserted Article 21A?	21A	38	0.534 vs. 0.614	242_0, 257A_0, 31D_0

A top-20 candidate pool would contain the relevant chunk for **6 of 8** of these queries, and a top-50 pool for **8 of 8**.

F. Latency

Setup: CPU; 52 queries × 3 runs (after warm-up); LLM: 120 saved calls.

Stage	Mean (ms)	Median (ms)	p95 (ms)
Bi-encoder retrieval (top 10)	23.0	21.6	31.4
Cross-encoder re-ranking (10 pairs)	482.0	506.1	595.8
Pipeline 1 retrieval total	23.0	21.6	31.4
Pipeline 2 retrieval total (retr + rerank)	505.0	527.6	621.5
LLM generation (Gemini, per call)	5147.8	4442.9	9910.5

- Re-ranking 10 chunks costs ∼21x the bi-encoder search over all 1035 chunks.
- Per chunk: bi-encoder ∼0.022 ms (1 query encoding + dot products), cross-encoder ∼48.2 ms (a full transformer pass per pair)

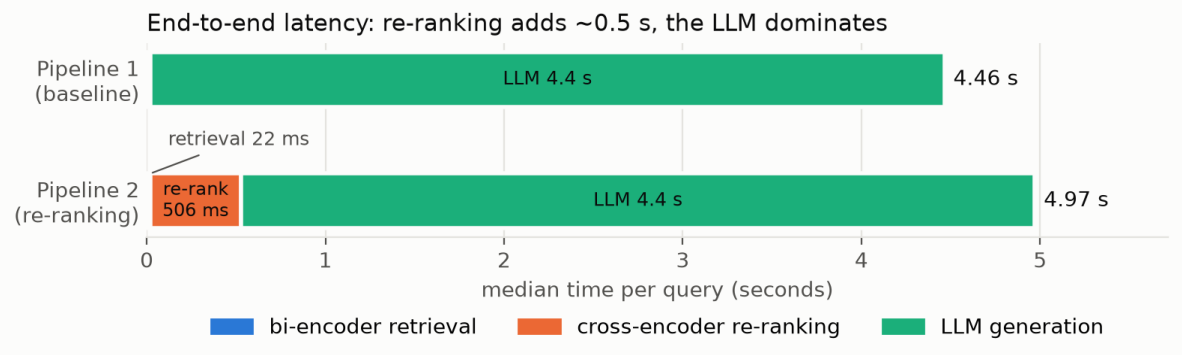


Figure 1: Median end-to-end time per query, split by stage.

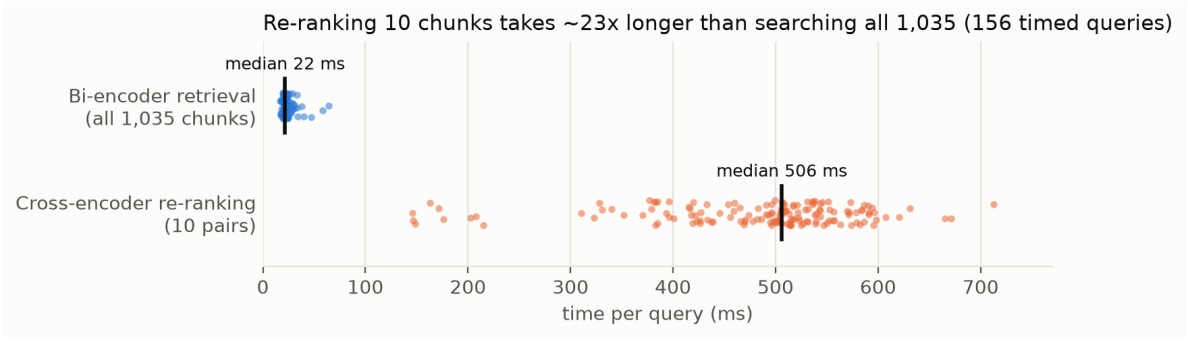


Figure 2: Bi-encoder retrieval vs. cross-encoder re-ranking, every timed query.